Final Project 2110215

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1 Overview

This application is a game called 'Elemental Slots'. It is based on a slot machine game where you spend money and get rewards based on your luck. Every roll is a random 6-digit number where it will be tested through each *achievement* and give handout if the condition is met.

Repository	https://github.com/2110215-ProgMeth/cp-project-2025-2-sanelarbkoi
JavaDoc	https://paristp.github.io/2110215_javadoc/

1.1 Menu

When you first open the game, you will be welcomed with this menu screen. A Vietnamese music will start playing in the background. There are 4 buttons you can press; START, SETTINGS, LEADERBOARD and EXIT.

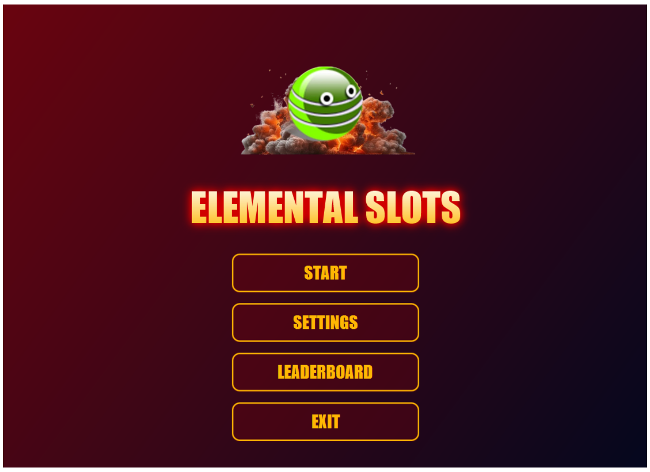


Figure 1.1: Menu screen

If you press START, you will see this alert. You can type your name into the field and press OK to log in.

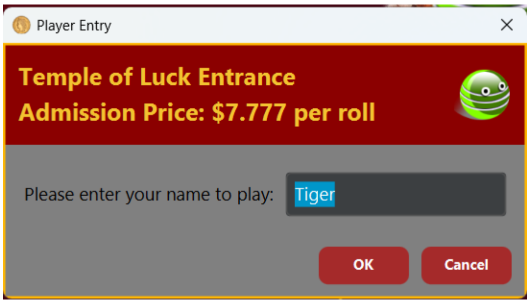


Figure 1.2: Name input alert

If you try to put in an empty name however, you will get this warning alert. You will have to type your name again to get into the game.

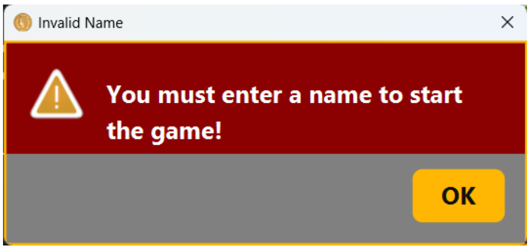


Figure 1.3: Invalid name alert

1.2 Settings

If you press SETTINGS, you will open settings page. Here, you can use these sliders to adjust the volume of game song and sound effects. You can go back to *menu screen* by hitting BACK.

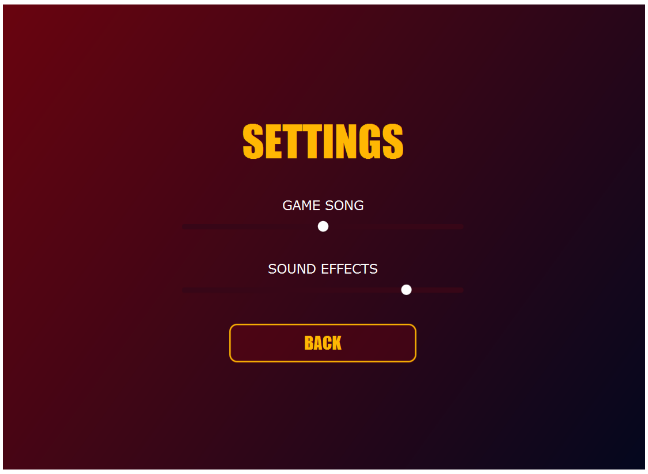


Figure 1.4: Setting page

1.3 Leaderboard

If you press LEADERBOARD, you will open temple leaderboard page. Initially it won't have any data but if after some people have played, their stats will be shown here.

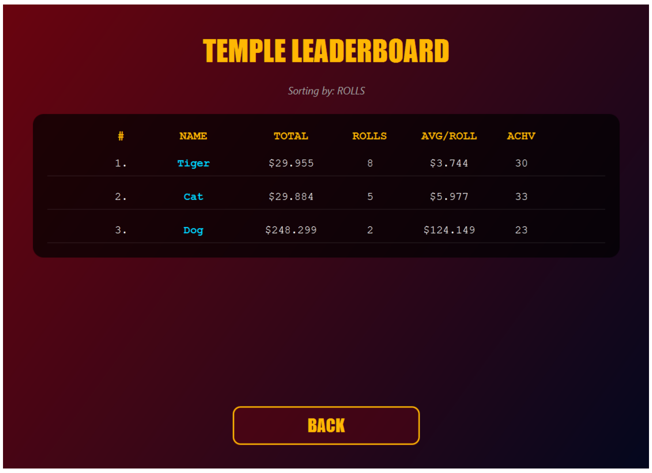


Figure 1.5: Leaderboard screen

You can press on each column to change the sorting condition. If you click on a player's name, you will open their *badge collection screen.*

1.4 Badge Collection

This is a collection page that shows all achievements that a player has gained. Unlocked badges will be shown in gray '?'. There are 138 badges in total from 6 rarities.

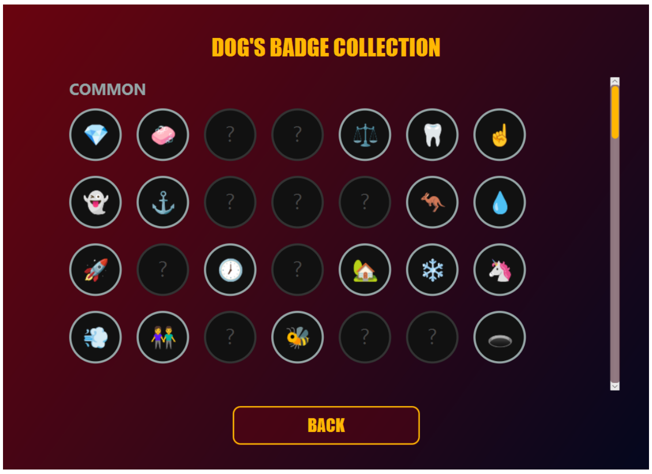


Figure 1.6: Collection

By clicking on a badge, it will show this pop-up screen. It contains player's record on getting this achievement.

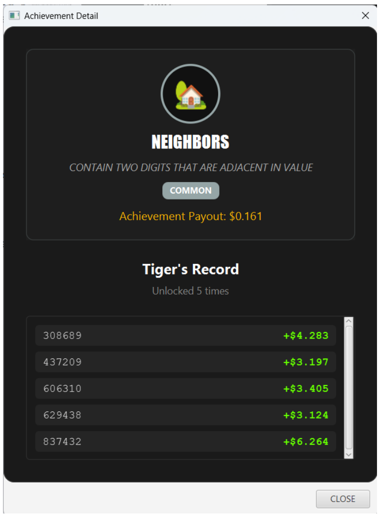


Figure 1.7: Collection

1.5 Loading Screen

If you are logging in for the first time, it will open this loading screen while the system is calibrating slot machine's stats. Every new player gets $200 to start. If done, it will open *game screen*.

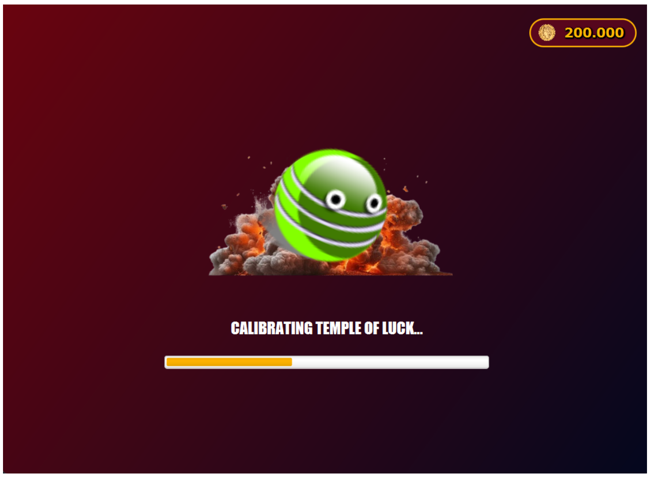


Figure 1.8: Loading screen

1.6 Game Screen

This screen is where the slot machine is located. You can play by clicking ROLL. It costs $7.777 and will roll a random number.

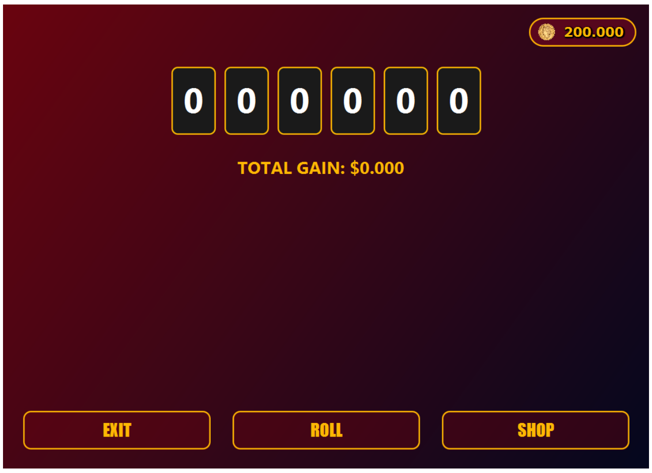


Figure 1.9: Initial game screen

Then, every achievement which the roll satisfies its condition will appear one-by-one in the middle. For example, if your roll is '410497', it will satisfy the 'Prime' achievement because it is a prime number.

When an achievement box shows up, it will reward you with its value based on how rare it is to get. You can read its name, condition, as well as its rarity. If you get this achievement for the first time, it will have a red 'new' tag. You can also press SKIP to jump to the end.

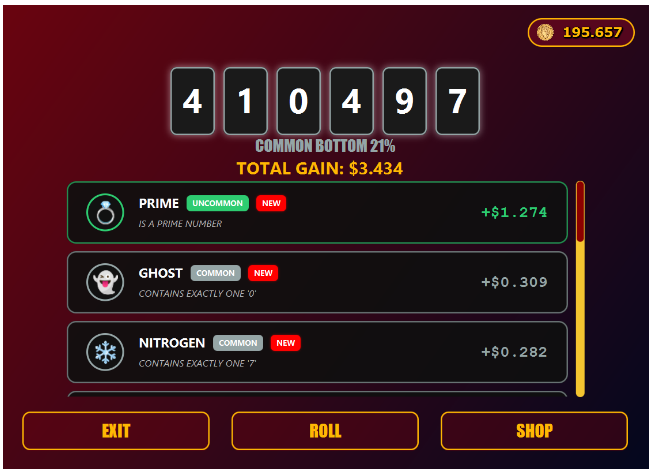


Figure 1.10: Roll result

After every achievement showed up, the game will tell you the roll's percentile and your total money gained. You can scroll through the details of the last roll. Every roll is saved to player's history and updated to *badge collection*.

1.7 Booster Shop

If you pressed SHOP, you will open booster shop. You will see offers of boosts showing names, condition, price and amount you have. Each boost will give you extra payout in percentages of your next roll if it satisfies the condition or else will be wasted.

Figure 1.11: Initial shop screen

You can click BUY to pay money and add it to your inventory. But if you don't have enough money, it will show 'Can't buy' instead. After you buy, you can click ACTIVATE to add it to your next roll.

Once you press ACTIVATE, it will take one of your stored item and add it to your active boost list. If you press the same button again, it will DEACTIVATE that item and bring it back to your inventory. Every active boosts will be used on the next roll.

Figure 1.12: Buying boosts

If you scroll down to bottom, you will see BUY ALL and ACTIVATE ALL buttons. Pressing BUY ALL will buy every boosts in the shop that you can afford except for *Krusty Kash*. Pressing ACTIVATE ALL will activate (or deactivate) any boosts you have.

Figure 1.13: Buy & activate all

Krusty Kash is a special boost that it gives you free money but will take 175% of payout on your next roll (kind of like a loan in case you fall short on money). Once bought it will automatically be activated.

1.8 Game Screen (cont.)

If you play a game with boosts equipped and some of it does give rewards, it will show result next to 'Total gain' text. There will be an I icon that you can press to see more information of boost bonus.

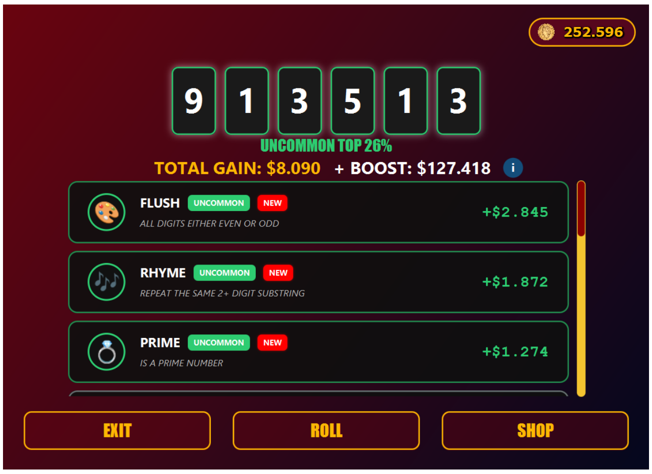


Figure 1.14: Roll result with boost

Figure 1.15: Boost breakdown alert

If you have less than $7.777, you will have to borrow money from Krusty Kash (if you can) or you will get this message once you click ROLL.

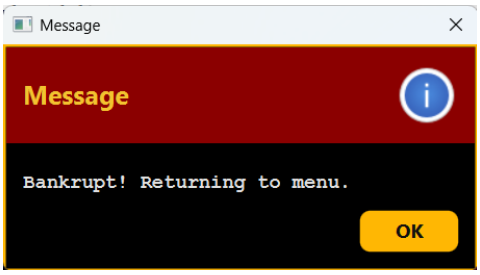
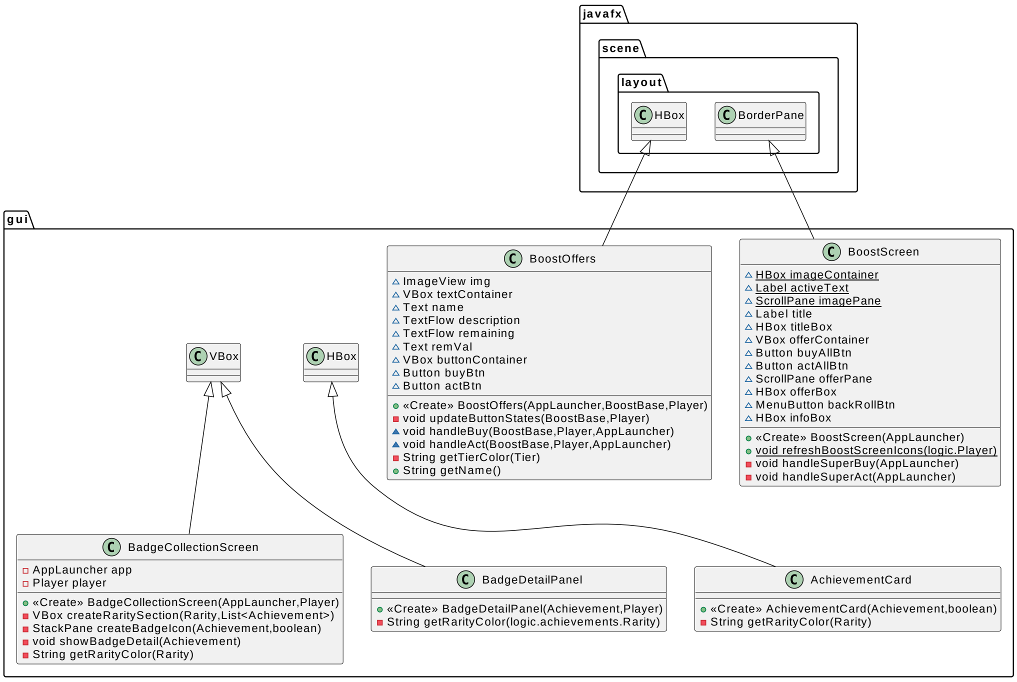


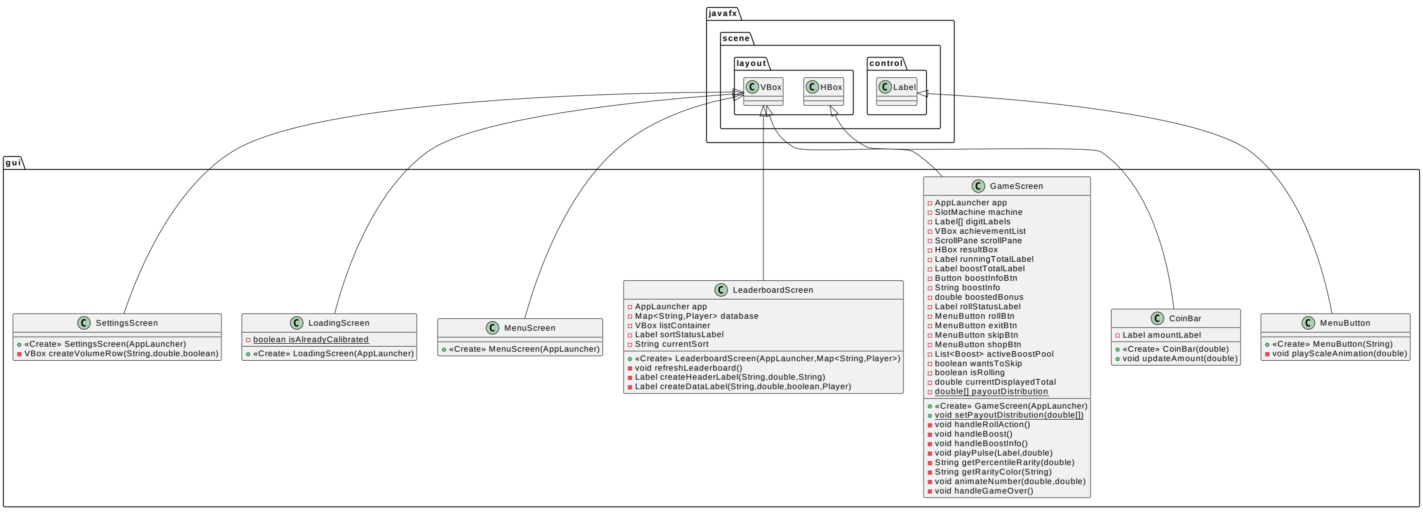
Figure 1.16: Bankrupt alert

You are bankrupt and will then be redirected to *menu screen* and your player data will be removed from the game. You can still create a new player with the same name but all past info will be removed.

2 UML Diagrams



(a) First half



(b) Second half

Figure 2.1: UML of package `gui`

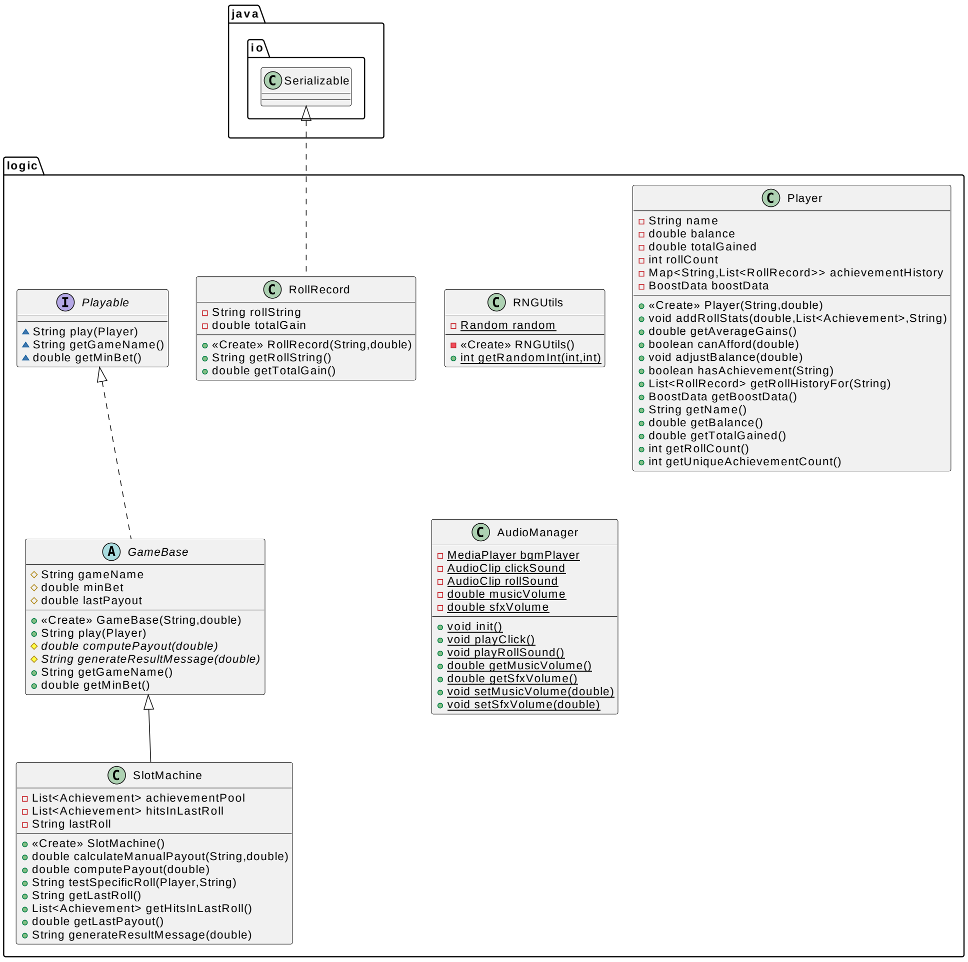


Figure 2.2: UML of package `logic`

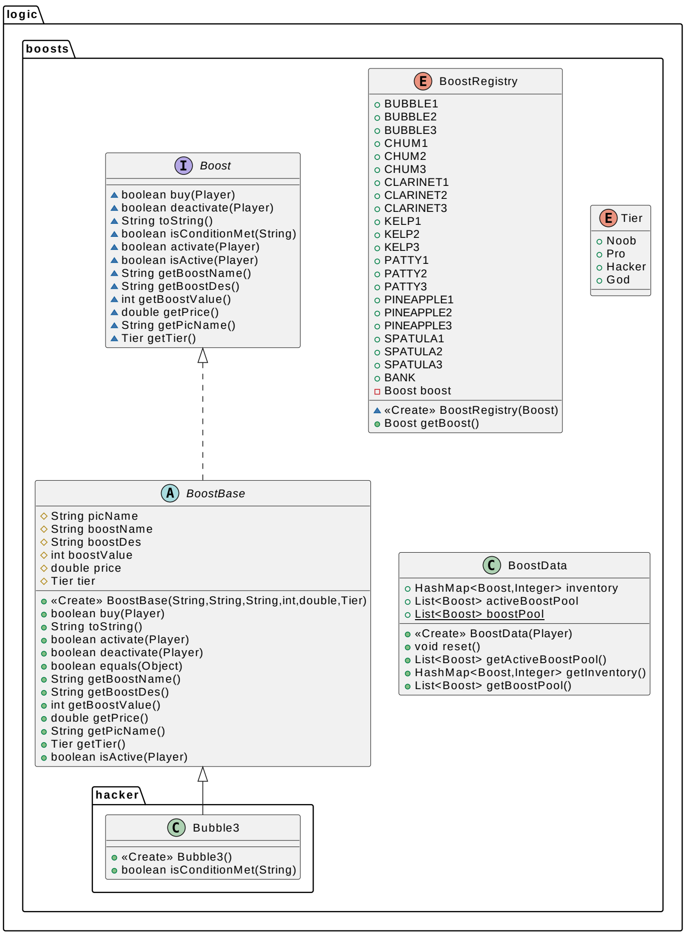


Figure 2.3: UML of package `logic.boosts`*

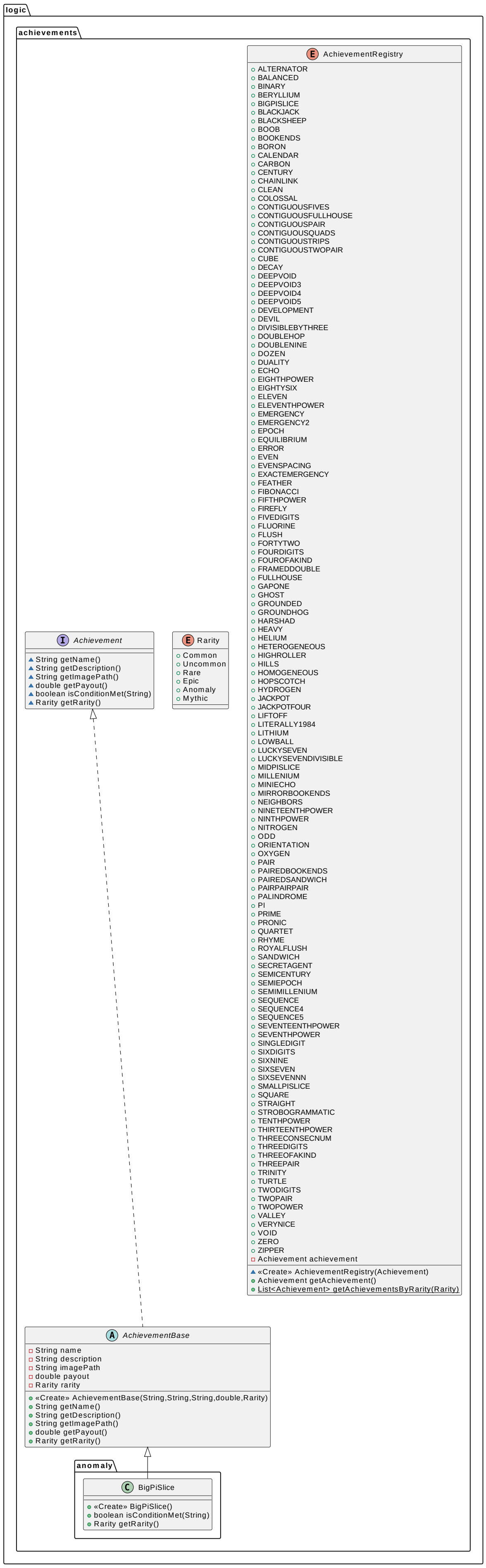


Figure 2.4: UML of package `logic.achievements`*

* These are classes that aren't included to the UML diagram because there are way too many to show. However, there is already an example class included to represent *boost* (`Bubble3`) and *achievement* (`BigPiSlice`) which the rest are all equivalent to.

`logic.achievements.anomaly`

1. `BigPiSlice`
2. `Binary`
3. `Epoch`
4. `Fibonacci`
5. `FifthPower`
6. `Homogeneous`
7. `Millenium`
8. `Pi`
9. `RoyalFlush`
10. `SemiEpoch`
11. `TwoDigits`
12. `TwoPower`

`logic.achievements.common`

1. `Beryllium`
2. `Boron`
3. `Carbon`
4. `ContiguousPair`
5. `Even`
6. `Fluorine`
7. `GapOne`
8. `Ghost`
9. `Grounded`
10. `Helium`
11. `Heterogeneous`
12. `Hills`
13. `Hopscotch`
14. `Hydrogen`
15. `LiftOff`
16. `Lithium`
17. `LuckySeven`
18. `LuckySevenDivisible`
19. `Neighbors`
20. `Nitrogen`
21. `Odd`
22. `Oxygen`
23. `Pair`
24. `Quartet`
25. `SixDigits`
26. `ThreeKind`
27. `TwoPair`
28. `Void`

`logic.achievements.epic`

1. `Boob`
2. `Colossal`
3. `ContiguousFives`
4. `Cube`
5. `Decay`
6. `DeepVoid4`
7. `Development`
8. `Echo`
9. `Emergency2`
10. `JackpotFour`
11. `Literally1984`
12. `MidPiSlice`
13. `PairPairPair`
14. `Palindrome`
15. `Pronic`
16. `SemiMillenium`
17. `Sequence5`
18. `SixSevennn`
19. `Straight`
20. `Strobogrammatic`
21. `ThreeConsecNum`
22. `ThreeDigits`
23. `VeryNice`
24. `Zipper`

`logic.achievements.mythic`

1. `DeepVoid5`
2. `EighthPower`
3. `EleventhPower`
4. `ExactEmergency`
5. `Groundhog`
6. `NineteenthPower`
7. `NinthPower`
8. `SeventeenthPower`
9. `SeventhPower`
10. `SingleDigit`
11. `TenthPower`
12. `ThirteenthPower`
13. `Zero`

`logic.achievements.rare`

1. `BlackSheep`
2. `Bookends`
3. `Calendar`
4. `Century`
5. `ContiguousFullHouse`
6. `ContiguousQuads`
7. `DeepVoid3`
8. `Devil`
9. `DivisibleByThree`
10. `DoubleNine`
11. `Duality`
12. `Emergency`
13. `Error`
14. `EvenSpacing`
15. `Firefly`
16. `FourDigits`
17. `FramedDouble`
18. `Heavy`
19. `Jackpot`
20. `MirrorBookends`
21. `Orientation`
22. `PairedBookends`
23. `PairedSandwich`
24. `SecretAgent`
25. `SemiCentury`
26. `Sequence4`
27. `SmallPiSlice`
28. `Square`
29. `ThreePair`
30. `Turtle`

`logic.achievements.uncommon`

1. `Alternator`
2. `Balanced`
3. `Blackjack`
4. `ChainLink`
5. `Clean`
6. `ContiguousTrips`
7. `ContiguousTwoPair`
8. `DeepVoid`
9. `DoubleHop`
10. `Dozen`
11. `EightySix`
12. `Eleven`
13. `Equilibrium`
14. `Feather`
15. `FiveDigits`
16. `Flush`
17. `FortyTwo`
18. `FourKind`
19. `FullHouse`
20. `Harshad`
21. `HighRoller`
22. `LowBall`
23. `MiniEcho`
24. `Nice`
25. `Peak`
26. `Prime`
27. `Rhyme`
28. `Sandwich`
29. `Sequence`
30. `SixSeven`
31. `Trinity`
32. `Valley`

`logic.boosts.hacker`

1. `Bubble3`
2. `Chum3`
3. `Clarinet3`
4. `Kelp3`
5. `Patty3`
6. `Pineapple3`
7. `Spatula3`

`logic.boosts.noob`

1. `Bubble1`
2. `Chum1`
3. `Clarinet1`
4. `Kelp1`
5. `Patty1`
6. `Pineapple1`
7. `Spatula1`

`logic.boosts.pro`

1. `Bubble2`
2. `Chum2`
3. `Clarinet2`
4. `Kelp2`
5. `Patty2`
6. `Pineapple2`
7. `Spatula2`